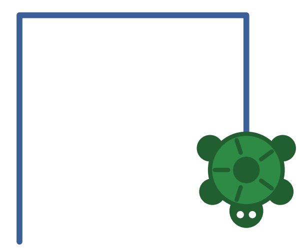
The Turtle Book

How to tell a turtle what to draw



This book belongs to ____________________

The Turtle Book
Updated September 5, 2026

What Is Inside

My Pictures

My Turtle Words

CHAPTER 1

Before You Start

Inside this computer there is a turtle, and the turtle will draw whatever you tell it to.

The turtle cannot hear you. It can only read. So you type words to it, and it does what the words say.

This book teaches you the words.

On the menu, pick “Draw Pictures with Logo Turtle” and press **Enter**.

The screen turns grey. In the middle is a small triangle. That is the turtle, seen from above. The point of the triangle is its nose, and it shows which way the turtle is facing.

Talking

`print [hi]`	write words at the bottom
`label [hi]`	write words on the drawing
`setlabelheight 40`	bigger letters
`random 30`	a surprise number from 0 to 29
`make "key readchar`	wait for a key and remember it
`wait 60`	pause for one second
`bye`	leave Logo

Again and Again

`repeat 4 [ ... ]`	do it again
`repcount`	which time this is, inside a repeat
`to name :size`	start teaching a word
`end`	finish teaching
`po "name`	show a lesson
`pots`	list every lesson
`erase "name`	forget a lesson
`save "mine.lg`	keep your lessons
`load "mine.lg`	get them back
`if :size > 150 [stop]`	stop rule

There is already a house on the screen. The turtle drew it while you were waiting. Later in this book you will find out how.

At the bottom of the screen is a question mark:

```
?
```

That is the turtle asking, “What should I do?”

Whatever you type goes after the question mark. When you press **Enter**, the turtle reads it and does it. Then the question mark comes back, ready for the next thing.

1.1 Big Letters or Small

This book writes the turtle’s words in small letters, like `fd 50`, because small letters are easier to type.

The turtle does not mind. `FD 50` and `fd 50` mean the same thing to it.

1.2 Leaving Logo

When you are finished, type:

```
bye
```

and press **Enter**. You go back to the menu.

If the turtle will not listen to you at all, hold down **Ctrl** and **Alt** and press **Backspace**. That always takes you back to the menu.

If you make a mistake
You will make lots of mistakes. Everybody does. The turtle never breaks. The worst thing that can happen is that it prints a message saying it did not understand. Chapter 13 tells you what the messages mean.

Moving

`fd 50`	forward
`bk 50`	back
`rt 90`	turn right
`lt 90`	turn left
`cs`	clear the screen, turtle to the middle
`home`	turtle to the middle, keep the drawing
`ht`	hide the turtle
`st`	show the turtle

The Pen

`pu`	pen up
`pd`	pen down
`setpc 4`	pen colour
`setbg 7`	screen colour
`setpensize [5 5]`	thicker pen
`fill`	pour colour into the shape you are inside

CHAPTER 14

Words the Turtle Knows

The turtle's words, all in one place.

CHAPTER 2

Meet the Turtle

The turtle knows four words for moving.

`fd`	forward
`bk`	back
`rt`	turn right
`lt`	turn left

Every one of them needs a number after it.

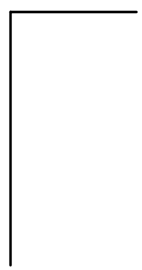
`fd 100` means "go forward 100 steps." The turtle's steps are tiny, so 100 of them is not very far.

`rt 90` means "turn right." The 90 says how much to turn. Chapter 3 is all about that number.

The turtle has a pen in its belly. Wherever it walks, it draws a line.

Type these three lines. Press **Enter** after each one.

```
fd 100
rt 90
fd 50
```



The turtle walked up, turned to face right, and walked a little way. The line it left behind is the drawing.

Now look at the turtle itself. Its nose points to the right. The turtle always remembers where it is and which way it is facing. `fd` and `bk` change where it is. `rt` and `lt` change which way it faces.

13.2 Square or “Square

`square` tells the turtle to draw a square.

`"square` tells the turtle you mean the name of the word.

That is why it is `po "square` and `erase "square`. You are talking about the word, not asking for the drawing.

If the turtle will not stop

Click **Logo** at the very top of the window, then click **Stop**.
If that does not work, hold down **Ctrl** and **Alt** and press **Backspace** to go back to the menu.

The turtle says	What it means
`I don't know how to squar`	A word the turtle has never learned. Usually it is spelled wrong, or it is a word you have not taught yet today.
`not enough inputs to fd`	`fd` wanted a number and did not get one.
`You don't say what to do with 50`	A number with no word in front of it. The turtle has a 50 and no idea what it is for.
`size has no value`	You used `:size` outside a lesson, so nobody handed the turtle a size.

13.1 When the Question Mark Does Not Come Back

If you type a `[` and forget the `]`, the turtle waits quietly for the rest. The question mark does not come back.

Type `]` and press **Enter**, and the turtle carries on.

2.1 Starting Over

Two more words:

`cs` clear the screen, and put the turtle back in the middle

`home` put the turtle back in the middle, but keep the drawing

`cs` is short for "clear screen." You will type it a lot.

If the turtle does not do it

The turtle wants every word spelled exactly right, with a space before the number. `fd 50` works. `fd50` does not, because the turtle cannot see where the word ends and the number begins.

Look at what you typed. Fix the one letter or the one space that is wrong, and press **Enter** again.

Try

- `bk 100`. Does the turtle draw when it walks backwards?
- `rt 90` four times in a row. Which way is it facing now?
- Draw the letter L. Then draw the letter T. You will need `bk` for the T.

CHAPTER 13

When Things Go Wrong

The turtle never breaks. It just tells you when it is confused.

When the turtle does not understand, it prints a message at the bottom of the screen. The message says exactly what confused it. These are the ones you will see most.

Each `if` line checks for one letter. If the key was `w`, go forward. If it was `a`, turn left. And so on.

The last line says `drive` again, so the turtle goes back to waiting for the next key. It keeps doing this until you press **Q**, which is the stop rule.

The picture above is one drive. Yours will look different, because you will press different keys.

> **If the keys do nothing**
> The turtle is checking for small letters. Make sure Caps Lock is off.

Try

- Make **W** go further. Make **A** and **D** turn less.
- Add a key that lifts the pen, and one that puts it down.
- Add a key that changes the colour.
- Add **C** for `cs`.

CHAPTER 3

Turning

The number after `rt` says how far round to turn.

`rt 90` is one corner. It is the turn the turtle makes at the corner of a square.

`rt 180` is two corners. The turtle ends up facing the opposite way, as if it turned round to look behind it.

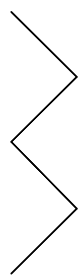
`rt 360` is four corners. That is all the way round. The turtle spins in a circle and ends up facing exactly the way it started.

Small numbers are small turns. `rt 45` is half a corner. `rt 10` is only a nudge.

`lt` works the same way, but the other direction.

Try this. Each turn here is half a corner, so the line zigs and zags.

```
rt 45
fd 50
lt 90
fd 50
rt 90
fd 50
lt 90
fd 50
```

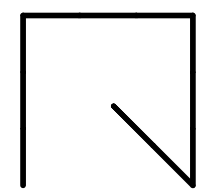


Try

- Change every `50` to `30`. Then to `80`.
- Change the `45` at the top to `20`. What changed?
- Make the zigzag longer by typing more lines.


```
to drive
  make "key readchar
  if :key = "w [fd 10]
  if :key = "s [bk 10]
  if :key = "a [lt 15]
  if :key = "d [rt 15]
  if :key = "q [stop]
  drive
end

drive
```



Here is how it works.

`readchar` waits for you to press one key, and hands that key over. `make "key` puts the key into a box called `key`. Then `:key` looks in the box. Those are the same two dots as `:size`, because they mean the same thing: “the thing I was given.”

CHAPTER 12

Driving the Turtle

This one is a game. You steer the turtle with the keys.

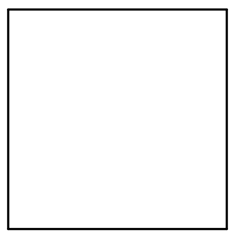
W	forward
S	back
A	turn left
D	turn right
Q	quit

CHAPTER 4

Saying It Again

Here is a square, the long way.

```
fd 100
rt 90
fd 100
rt 90
fd 100
rt 90
fd 100
rt 90
```



Look at what you typed. It is the same two lines, four times over.

The turtle has a word for “do this again.” It is `repeat`. After `repeat` you say how many times. Then you put the things to repeat inside square brackets: `[` and `]`.

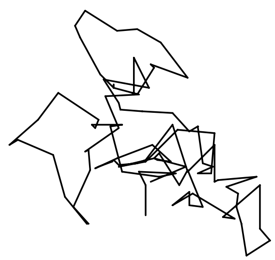
Here the colour is a surprise too. `random 6` gives 0 to 5, and adding 1 makes it 1 to 6. That skips black, which would not show up well.

Try

- Run `repeat 100 [wander]` three times. Do you get the same picture?
- `repeat 20 [square random 100]`
- `repeat 9 [house random 90 rt 40]`

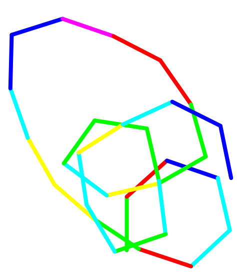
```
to wander
  fd random 30
  rt random 360
end

repeat 100 [wander]
```

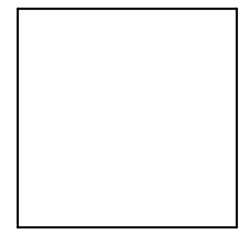


Every step is a surprise length, and every turn is a surprise turn.

```
setpensize [3 3]
repeat 30 [setpc 1 + random 6 fd 40 rt random 90]
```




```
repeat 4 [fd 100 rt 90]
```



This draws the very same square. The brackets show the turtle where the repeated part starts and stops.

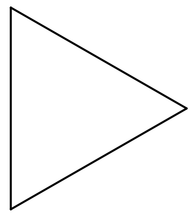
4.1 The Magic Number

A square has four corners, and the turtle turns `90` at each one. Four times 90 is 360. That is all the way round, which is why the turtle ends up facing the same way it started.

Every closed shape works like this. The turns always add up to 360.

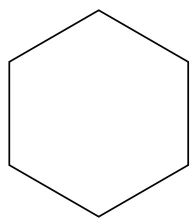
A triangle has three corners. Three turns that add up to 360 are 120 each.

```
repeat 3 [fd 100 rt 120]
```



A hexagon has six corners. Six turns that add up to 360 are 60 each.

```
repeat 6 [fd 60 rt 60]
```



Here is the whole table. The two numbers in each row always multiply to make 360.

CHAPTER 11

Surprises

The turtle can pick a number by itself.

`random 30` gives you a surprise number. It might be 0. It might be 29. It might be anything in between. It is never 30 itself. The number you give `random` is one more than the biggest it will pick.

Type `print random 10` five times. You get a different number nearly every time.

A drawing with `random` in it comes out different every time you draw it.

```
setpensize [3 3]
setpc 1
setlabelheight 40
label [HI THERE]
```

HI THERE

Put your own name in the brackets.

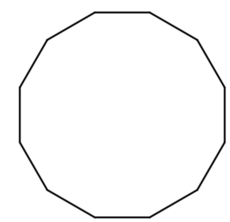
Try

- `print 100 * 100`
- `rt 90 label [sideways]`
- Write your name at the four corners of a square:
  `repeat 4 [label [ME] fd 100 rt 90]`

sides	turn	name
3	120	triangle
4	90	square
5	72	pentagon
6	60	hexagon
8	45	octagon
12	30	

The last row has no name. Guess what it will look like before you try it.

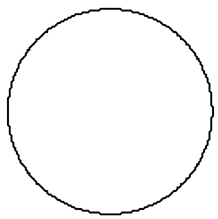
```
repeat 12 [fd 30 rt 30]
```



4.2 Round

Twelve sides is nearly round. What about 360 sides? The turtle takes 360 tiny steps and 360 tiny turns, and the corners disappear.

```
repeat 360 [fd 1 rt 1]
```



Try

- `repeat 4 [fd 100 rt 90]` but change `90` to `91`. Then `repeat 40` of that.
- `repeat 5 [fd 100 rt 144]` draws a star. Why does 144 work when a pentagon needs 72?
- `repeat 360 [fd 2 rt 1]` is a bigger circle.


```
print [Hello!]
print sum 2 3
print 7 * 6
print [I am a turtle.]
```

```
Hello!
5
42
I am a turtle.
```

10.1 Writing on the Picture

`label` writes on the drawing instead, right where the turtle is standing, facing the way the turtle is facing.

`setlabelheight` makes the letters bigger.

CHAPTER 10

The Turtle Can Talk

The turtle can write words as well as draw lines.

`print` writes at the bottom of the screen, where you type. Words go inside square brackets.

The turtle can do sums too. Give `print` a sum instead of words, and it works out the answer first. The star `*` means times.

CHAPTER 5

The Pen

The turtle can lift its pen off the paper.

`pu`	pen up
`pd`	pen down

With the pen up, the turtle still walks, but it leaves no line. Put the pen down and it draws again.

That is how you draw two things that do not touch each other.

```
repeat 8 [fd 15 pu fd 10 pd]
```


5.1 Colours

`setpc` sets the pen colour. It is short for “set pen colour.” Every colour has a number.

`setpensize` makes the line thicker. It wants two numbers in brackets. Keep them the same, like `[5 5]`.

If the turtle will not stop

Sometimes a program goes on and on. Move the mouse to the very top of the window and click on the word **Logo**. A little list drops down. Click **Stop**.
The question mark comes back, and the turtle is ready for you again.
If that does not work, hold down **Ctrl** and **Alt** and press **Backspace**. That leaves Logo and goes back to the menu.

Try

- In `spiral`, change `rt 90` to `rt 91`. Then to `rt 89`.
- Change `+ 10` to `+ 3`.
- Change `150` to `300`.
- Teach a `hexspiral` that turns `rt 60`.

9.1 The Stop Rule

Now look at the second line: `if :size > 150 [stop]`.

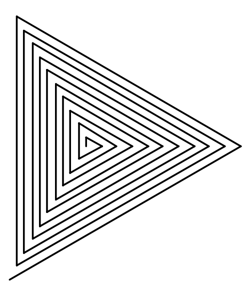
This is the stop rule. It says “if the number is bigger than 150, stop here.” Without it, the turtle would never stop. It would keep making bigger and bigger sides for ever.

Every word that uses itself needs a stop rule. Always write the stop rule first.

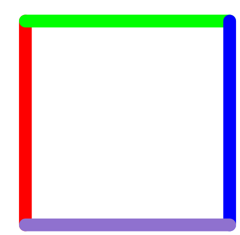
Here is the same trick with triangles.

```
to trispiral :size
  if :size > 150 [stop]
  fd :size
  rt 120
  trispiral :size + 5
end

trispiral 5
```




```
setpensize [5 5]
setpc 4
fd 80
rt 90
setpc 2
fd 80
rt 90
setpc 1
fd 80
rt 90
setpc 13
fd 80
rt 90
```



0	black	8	brown
1	blue	9	tan
2	green	10	forest green
3	light blue	11	aqua
4	red	12	salmon
5	magenta	13	purple
6	yellow	14	orange
7	white	15	grey

`setbg` changes the colour of the whole screen. It uses the same numbers. The screen starts out grey, which is 15, with a black pen, which is 0.

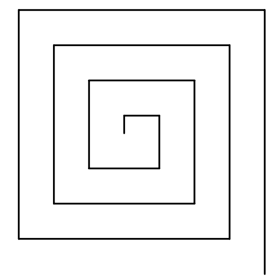
5.2 Counting Inside a Repeat

Inside a `repeat`, the turtle keeps count of which time it is on. The word for that count is `repcount`. The first time round, `repcount` is 1. The second time it is 2. And so on.

If you use `repcount` as the pen colour, every line gets a new colour.


```
to spiral :size
  if :size > 150 [stop]
  fd :size
  rt 90
  spiral :size + 10
end

spiral 10
```



Look at the last line of the lesson. `spiral` uses `spiral`. The word calls itself.

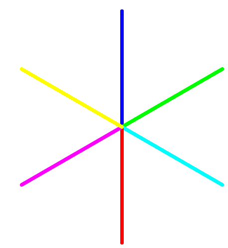
Here is what happens. `spiral 10` walks forward 10 and turns a corner. Then it says `spiral 20`. That walks forward 20 and turns a corner. Then it says `spiral 30`. Each side is a little longer than the last, so the line winds outwards.

CHAPTER 9

A Word That Uses Itself

This is the strangest trick in the book, and the best.


```
setpensize [3 3]
repeat 6 [setpc repcount fd 100 bk 100 rt 60]
```

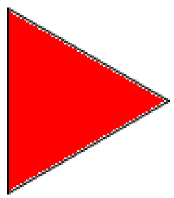


5.3 Filling In

`fill` pours colour into a shape.

First draw the shape. Then lift the pen and walk the turtle inside. Then say `fill`. The colour spreads out until it reaches the lines.

```
repeat 3 [fd 100 rt 120]
pu
rt 30
fd 30
pd
setpc 4
fill
```



If the turtle is outside the shape, the colour goes outside instead. So make sure it is standing inside first.

Try

- A dotted circle:
  `repeat 36 [fd 5 pu fd 5 pd rt 10]`
- A rainbow hexagon:
  `repeat 6 [setpc repcount fd 60 rt 60]`
- Type `cs`. The house comes back when you open Logo again. Fill it in.

Between the houses, the pen is up, so the turtle leaves no line on the way to the next one.

Try

- A street of five houses. Do they fit? If not, make them smaller.
- A door. It is a small tall rectangle. Where does the turtle have to walk before it draws it?
- A window. A chimney. A sun in the sky, using the circle from Chapter 4.

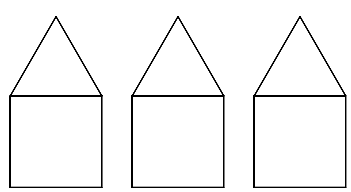
```
to square :size
  repeat 4 [fd :size rt 90]
end

to triangle :size
  repeat 3 [fd :size rt 120]
end

to house :size
  square :size
  fd :size
  rt 30
  triangle :size
  lt 30
  bk :size
end

to street
  repeat 3 [house 60 pu rt 90 fd 80 lt 90 pd]
end

street
```



CHAPTER 6

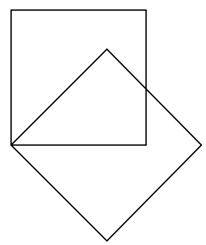
Teaching the Turtle a New Word

The turtle only knows the words it was born with. But you can teach it more.

Teaching a word is like giving a lesson. `to square` says "here is a lesson about a word called square." The lines after it are what the word means. `end` says the lesson is over.

```
to square
  repeat 4 [fd 100 rt 90]
end

square
rt 45
square
```



While you are teaching, the question mark goes away. Instead you see this:

```
>
```

That is the turtle saying "I am listening to the lesson. I will not do these things yet."

After you type `end`, the question mark comes back.

Now `square` is a word the turtle knows, just like `fd` or `rt`. Type it and the turtle draws a square. Turn the turtle and type it again for a tilted square.

Read the lesson for `house` one line at a time, and follow the turtle with your finger.

First it draws the square. Then `fd :size` walks it up the left wall, to the top corner. Then `rt 30` tilts it a little, so the triangle leans over the square instead of standing up beside it. It draws the triangle. Then `lt 30` and `bk :size` walk it back down to where it started.

This is the house that is on the screen when you open Logo. Now you know how it got there.

8.1 A Street

The last two lines of `house` matter. They put the turtle back exactly where it began, facing the same way.

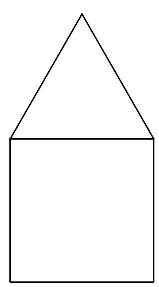
That means you can draw a house, walk to the right, and draw another one.

```
to square :size
  repeat 4 [fd :size rt 90]
end

to triangle :size
  repeat 3 [fd :size rt 120]
end

to house :size
  square :size
  fd :size
  rt 30
  triangle :size
  lt 30
  bk :size
end

house 100
```



6.1 Words About Words

`po "square`	show the lesson for square
`pots`	list every word you have taught
`erase "square`	make the turtle forget square

Look at the quote mark in `po "square`. It is only at the front, with nothing at the end. That looks strange, but it is right.

The quote mark tells the turtle "I mean the name square, not the drawing." Without it, the turtle would draw a square instead of showing you the lesson.

6.2 Remembering

When you leave Logo, the turtle forgets every word you taught it.

To keep them, type:

```
save "mine.lg
```

The turtle writes all your words down under the name `mine.lg`.

Next time you open Logo, type:

```
load "mine.lg
```

and every word comes back.

Try

- Teach the turtle `triangle`. Then `hexagon`.
- `repeat 8 [square rt 45]`
- Teach it `box`: a square, and then walk back to where it started, facing the same way. Check it with `repeat 3 [box fd 120]`.

CHAPTER 8

Building Big Things

Big pictures are made of small words.

A house is a square with a triangle on top. You already know how to teach both of those. The tricky part is getting the roof to sit flat.

`flower` draws twelve squares, turning a little between each one. It hands its own number on to `square`, so a big flower is made of big squares.

Twelve turns of 30 make 360. That is the magic number again, which is why the flower comes all the way round.

Try

- `flower 30`. Then `flower 120`.
- In `flower`, change `12` to `6` and `30` to `60`. Then try numbers that do not multiply to 360, and see.
- Teach `triangle :size` and make a flower out of triangles.

#

CHAPTER 7

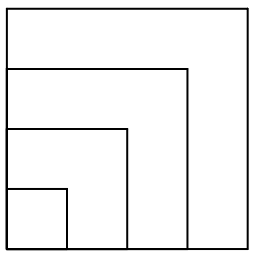
Words That Take a Number

`fd` takes a number. Why can't `square`?

It can.

```
to square :size
  repeat 4 [fd :size rt 90]
end

square 30
square 60
square 90
square 120
```



Look at the first line: `to square :size`.

The `:size` part means "square wants a number, and inside the lesson we will call it size."

The two dots in front of `size` are important. They mean "the number that was handed to me." So `fd :size` means "go forward by whatever number I was given."

Then `square 30` draws a small square, and `square 120` draws a big one. It is the same lesson every time, with a different number.

7.1 Words Inside Words

A word you taught the turtle can be used inside another lesson.

```
to square :size
  repeat 4 [fd :size rt 90]
end

to flower :size
  repeat 12 [square :size rt 30]
end

flower 80
```